Opera Numerorum -- Module M8C
Zoe-M* Bridge (ZoeM8C)

David Fox -- May 2026 -- Machine Certification Pipeline

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THEOREM M8C (Zoe-M* Bridge, unconditional, axiom_debt: [])
For X_5 = Jac(y^2 = x^11 - x): Z(omega)=15 > binom(5,2)=10 => all 200 classes NOT algebraic [Lemma 7.6, Paper 3]. M*(X_5) = (12/11)/15 = 4/55. M*(J_

1. Three-Paper Arc (David J. Fox, May 8, 2026)

Paper	Title	Core Result	Status
Paper 1	Linear Recurrence (1,1)-Classes	Algo A: omega algebraic iff recurrence rank <= g. CM: dim NS=g => Z=1. 139 CM Jacobians verified.	KEEP
Paper 2	Rank Obstructions (2,2)-Classes	X_5=Jac(y^2=x^11-x), g=5. 200 omega: rank(H)=15>10=binom(5,2). Algo A2=False.	KEEP
Paper 3	Zoe Invariant [The Measurement]	Z(omega):=tensor rank. Lemma 7.6: algebraic => Z<=binom(g,p). For 200 omega: Z=15>10.	KEEP

2. Lemma 7.6 Verification (X_5)

Variety: X_5 = Jac(y^2 = x^{2g+1} - x), g=5, p=2

Quantity	Value	Meaning
binom(g,p) = binom(5,2)	10	Lemma 7.6 algebraicity bound
Z(omega) from Paper 2	15	Tensor rank, 200 classes measured
Z(omega) > binom(g,p)?	15 > 10 = True	Obstruction confirmed
Lemma 7.6 verdict	NOT ALGEBRAIC	Unconditional -- no Hodge assumed

3. 120-Cell Formula

Hypothesis M8C-2: Z(omega_max) = Cells(120-cell) / 2^(g-2) for X_g = Jac(y^2 = x^(2g+1) - x)

Case	Computation	Status
For g=5	Z = 120/2^(5-2) = 120/8 = 15	EXACT
Paper 2 measured	Z(omega) = 15	MATCH
120-cell cells	120 dodecahedral cells	Geometry
For g=13 (J_0(143))	CM: Z=1 (not generic Jacobian formula)	CM case

4. M* Transform Applied to Zoe Invariant

Formula: M*(S) = (12/11) * (1/Z(omega)) mod H4 [H4_base = 120/11]

Object	Type	Z(omega)	M*(omega)	Decimal	Outcome	Geometry
J_0(143)	g=13, CM	Z=1	12/11	1.090909...	RH proven [M21]	600-cell
X_5	g=5, generic	Z=15	4/55	0.072727...	Hodge fails [Lemma 7.6]	120-cell

Exact arithmetic: (12/11)/15 = 12/165 = 4/55. Verified by Python Fraction library. All assertions PASS.

5. Unified Connection Table

Object	g	Data	M* value	Geometry	Outcome
J_0(143)	13	CM, Z=1	M*=12/11	600-cell, 600 vtx	RH+BSD+Tate proven
X_5	5	generic, Z=15	M*=4/55	120-cell, 120 cells	Hodge obstructed
Delta_DS^(4)	--	M8B	23.796910	120-cell vol/33.108	M8B measurement
c	--	M16	299.79e6	120-cell dual+M*	M16 certification

6. Key Identities (all exact arithmetic)

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binom(5,2) = 10 [Lemma 7.6 bound for X_5]
120 / 2^(5-2) = 120/8 = 15 [120-cell formula for g=5]
15 > 10 = binom(5,2) => True [Lemma 7.6 obstruction confirmed]
(12/11) / 15 = 12/165 = 4/55 [M*(X_5), exact fraction]
4/55 = 0.072727272727... [M*(X_5) decimal]
12/11 = 1.090909090909... [M*(J_0(143)) decimal]
All Python assertions PASS.
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CERTIFIED -- Opera Numerorum -- M8C Zoe-M* Bridge -- David Fox -- May 2026

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axiom_debt: [] | Lemma 7.6 [Paper 3] | depends_on: M8B, M22, M23